“NextGen Attendance Management System using Facial Recognition”

A Project report submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING (CSE)

With

Artificial Intelligence & Machine Learning

Submitted to

SANDIP UNIVERSITY, SIJOUL

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CERTIFICATE

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DECLARATION

I hereby declare that the project report entitled **"NextGen Attendance Management System using Facial Recognition"** submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (CSE) With Artificial Intelligence and Machine Learning** my original work carried out under the guidance of **Dr. Syed Anas Ansar.**

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APPROVAL CERTIFICATE

The project report entitled **"NextGen Attendance Management System using Facial Recognition"**, submitted by **"Kaushal Kumar Ray"** bearing **PRN No: 240205241023**, has been examined by us and is hereby approved for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (CSE)** at **Sandip University**.

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ABSTRACT

Traditional attendance systems are inefficient, time-consuming, and prone to manipulation such as proxy attendance, which reduces reliability and increases administrative effort. With the rapid advancement of deep learning and computer vision, facial recognition has emerged as a robust and secure alternative for identity verification. This project presents an intelligent attendance management system based on deep learning techniques and facial recognition models.

The proposed system captures facial images of students and processes them using advanced models such as ArcFace and DeepFace to extract high-dimensional facial embeddings. These embeddings represent unique facial features and are compared with stored data using similarity metrics to accurately identify individuals. The system is designed to handle variations in lighting, pose, and facial expressions, thereby improving recognition performance compared to traditional methods.

A web-based application developed using the Flask framework enables real-time attendance tracking, monitoring, and visualization. The backend is integrated with a PostgreSQL database to ensure secure and structured data storage, while Cloudinary is used for efficient image storage and management. The system also incorporates additional features such as attendance percentage calculation and an automated leave management module based on predefined criteria.

Experimental observations indicate that the proposed system significantly improves accuracy, reduces manual effort, and enhances overall system security. Furthermore, the modular architecture and scalable design make it suitable for deployment in educational institutions and organizations.

Keywords: Face Recognition, Deep Learning, Attendance System, ArcFace, DeepFace, Computer Vision**.**

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Chapter 1

Introduction

1.1 Background

Attendance management is a fundamental administrative activity in educational institutions and organizations, directly influencing performance evaluation, discipline monitoring, and compliance tracking. Traditionally, attendance has been recorded manually through roll calls or maintained using paper-based registers. While simple, these methods are inefficient, time-consuming, and prone to human error and manipulation.

With the advancement of digital technologies, alternative systems such as RFID (Radio Frequency Identification), biometric fingerprint scanners, and smart cards have been introduced. However, these systems still present limitations such as hardware dependency, risk of proxy usage, hygiene concerns (in touch-based systems), and additional infrastructure costs.

Recent developments in deep learning, particularly in the field of computer vision, have enabled significant improvements in facial recognition systems. Techniques based on Deep Convolutional Neural Networks (DCNNs) can extract high-dimensional facial features and generate unique embeddings for accurate identity verification. These advancements make facial recognition a reliable, contactless, and scalable solution for automated attendance management.

This project proposes a web-based automated attendance system that leverages facial recognition technology to detect and identify individuals in real-time, thereby streamlining attendance processes while enhancing accuracy and security.

1.2 Problem Statement

Traditional attendance systems are inefficient, labor-intensive, and susceptible to manipulation such as proxy attendance. Manual processes require significant time for recording and maintaining attendance data, often leading to inaccuracies and delays in record management.

Existing digital systems, while improving certain aspects, still depend on physical interaction or external devices, which may be inconvenient, costly, or vulnerable to misuse.

Therefore, there is a need for a **secure, efficient, and automated attendance system** that can:

- Eliminate proxy attendance
- Reduce manual effort
- Ensure accurate and real-time data recording
- Operate in a contactless and scalable manner

1.3 Objectives

The primary goal of this project is to design and develop a smart attendance management system using facial recognition technology. The system is intended to improve efficiency, accuracy, and security compared to traditional methods.

The key objectives of the project include:

- To develop an automated attendance system using deep learning-based facial recognition
- To reduce manual effort involved in attendance recording and management
- To eliminate proxy attendance by ensuring identity verification through facial features
- To enhance recognition accuracy using advanced deep learning models
- To securely store and manage attendance data in a centralized database
- To design a user-friendly web interface for seamless interaction between users and administrators

1.4 Scope of the Project

The proposed system focuses on automating the attendance process through facial recognition within a web-based environment. It is designed to operate in real-time and provide accurate attendance tracking with minimal human intervention.

The scope of the system includes:

- Real-time face detection and recognition using a camera or webcam
- Automated marking of attendance upon successful identification

- Secure storage and retrieval of attendance data using a database system
- Web-based platform for accessing attendance records, reports, and user management features
- Administrative control for monitoring attendance and managing users

While the system aims to be robust and efficient, certain limitations exist. The performance of facial recognition may be affected by environmental factors such as lighting conditions, camera quality, and facial obstructions. Additionally, the system requires initial dataset collection and model training to achieve optimal accuracy.

1.5 Organization of the Report

This report is structured to provide a comprehensive understanding of the proposed system, from conceptual foundations to implementation and evaluation.

- **Chapter 1: Introduction**
  Provides an overview of the background, problem statement, objectives, and scope of the project
- **Chapter 2: Existing System and Literature Review**
  Discusses current systems, their limitations, and reviews existing research and technologies related to facial recognition
- **Chapter 3: System Overview**
  Explains system analysis, requirements, and the overall working environment including hardware and software requirements
- **Chapter 4: System Design**
  Describes the system architecture, data flow diagrams, use case diagrams, database design, and methodology
- **Chapter 5: System Implementation**
  Presents the frontend, backend, database implementation, and key features of the system
- **Chapter 6: System Testing**
  Covers testing strategies, test cases, and performance evaluation of the system
- **Chapter 7: Results and Discussion**
  Shows system outputs, performance analysis, and discussion of results
- **Chapter 8: Conclusion and Future Work**
  Summarizes the outcomes of the project and suggests possible future improvements

- **Appendix**

  Includes code snippets, additional screenshots, and API documentation

Chapter 2

Existing System and Literature Review

2.1 Existing System

Attendance management in most educational institutions is still largely dependent on traditional or semi-digital systems. The most common approach involves manual roll calls or maintaining physical registers, where attendance is recorded by instructors during class hours. While this method is simple, it is inefficient and consumes valuable instructional time.

In recent years, some institutions have adopted digital solutions such as RFID-based systems, biometric fingerprint scanners, and smart card technologies. These systems aim to automate attendance recording; however, they still rely heavily on physical interaction and additional hardware infrastructure. Moreover, they are not entirely secure, as RFID cards can be shared and fingerprint systems may face issues related to hygiene and device dependency.

The major problems observed in existing systems include:

- Long time required for attendance marking
- Dependency on manual or semi-manual processes
- Possibility of proxy attendance (e.g., card sharing)
- Lack of real-time tracking and analytics
- Increased administrative workload

2.2 Limitations of Existing System

Despite technological advancements, current attendance systems continue to face several limitations that reduce their effectiveness and scalability.

These limitations can be summarized as follows:

- ***Slow Processing:*** Manual and device-based systems take time for each individual, making them inefficient in large classrooms
- ***Proxy and Manipulation:*** Students can bypass systems using proxies (e.g., RFID sharing or manual marking)

- ***<u>Hardware Dependency:</u>*** Systems require dedicated devices such as scanners, which increase cost and maintenance
- ***<u>Lack of Flexibility:</u>*** Limited adaptability to remote or hybrid environments
- ***<u>Poor Data Management:</u>*** Many systems still rely on fragmented or paper-based record keeping
- ***<u>Hygiene Issues:</u>*** Touch-based biometric systems are not ideal in health-sensitive environments

These challenges highlight the need for a more advanced, automated, and contactless attendance solution.

2.3 Proposed Solution Overview

To overcome the limitations of existing systems, this project proposes a deep learning-based automated attendance system using facial recognition integrated into a web application.

The proposed system leverages computer vision and deep learning techniques to detect and recognize faces in real-time using a camera feed. Once a face is identified, the system automatically marks attendance and stores the data securely in a centralized database. This eliminates the need for manual intervention and significantly reduces the possibility of proxy attendance.

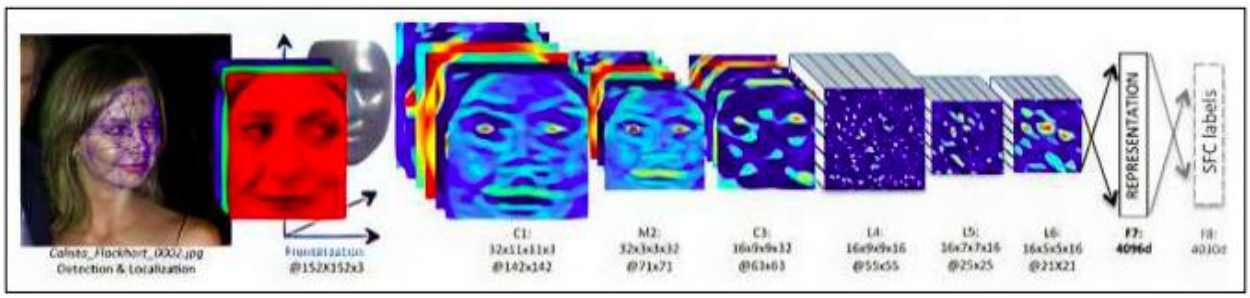

Figure 1 Outline of the DeepFace Architecture

Key features of the proposed system include:

- Contactless and automated attendance marking
- Real-time face detection and recognition
- High accuracy using deep learning models

- Secure storage of attendance records
- Web-based interface for accessibility and management
- Scalability for large classrooms or institutions

This approach ensures improved efficiency, accuracy, and user experience compared to traditional systems.

2.4 Literature Review

Facial recognition technology has evolved significantly over the years, transitioning from traditional machine learning techniques to advanced deep learning-based approaches. Early face recognition systems relied on algorithms such as *Haar Cascade* for face detection and *Local Binary Patterns Histogram (LBPH)* for recognition. While these methods were computationally efficient, they suffered from low accuracy, especially under varying lighting conditions, facial expressions, and occlusions.

With the advancement of deep learning, modern systems now utilize *Deep Convolutional Neural Networks (DCNNs)* to extract high-level facial features. Models such as *FaceNet* and *DeepFace* introduced the concept of mapping facial images into a high-dimensional embedding space, where similarity between faces is measured using distance metrics. These approaches significantly improved recognition accuracy and robustness.

Recent research emphasizes the effectiveness of embedding-based models, where each face is represented as a numerical vector. These embeddings enable efficient comparison between faces using similarity measures such as cosine distance. As a result, deep learning-based systems outperform traditional methods in terms of accuracy, scalability, and real-world applicability.

A major advancement in this domain is the introduction of *ArcFace*, which improves recognition performance by incorporating an additive angular margin loss. This approach enhances the discriminative power of facial features by increasing inter-class differences (between different individuals) and reducing intra-class variations (within the same individual). According to the study provided in the uploaded document, ArcFace achieves state-of-the-art performance even in challenging scenarios such as disguised face recognition.

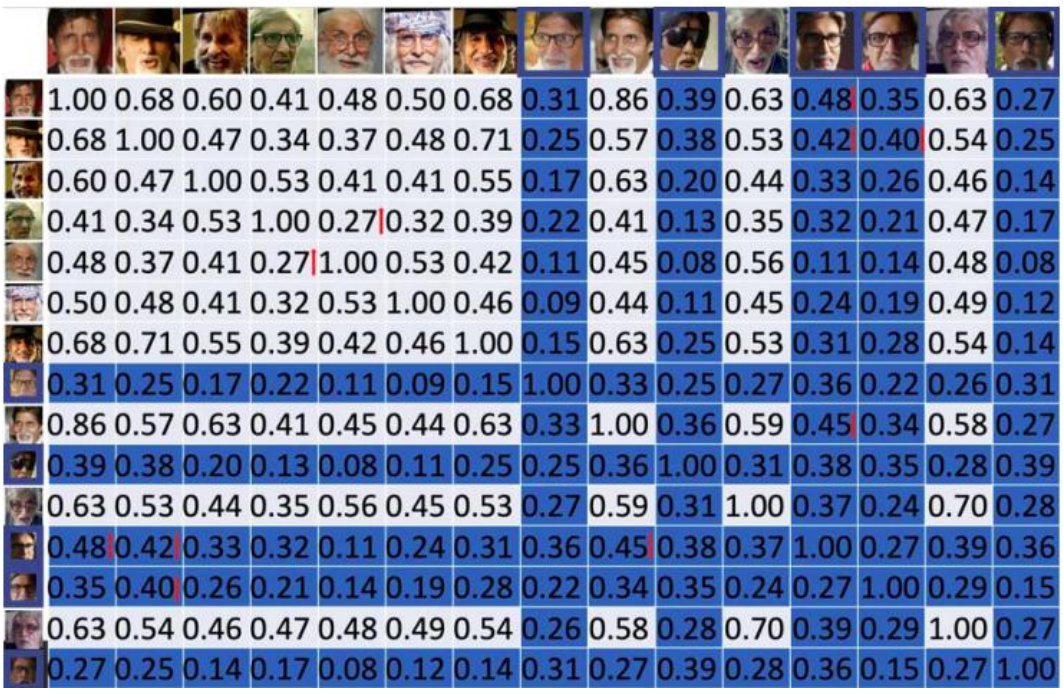

Figure 2 Cosine Similarity Matrix by ArcFace

The paper further highlights that facial recognition becomes particularly difficult when there are high intra-class variation (e.g., due to aging, makeup, or occlusion) and low inter-class variation (e.g., impersonation). ArcFace addresses this by generating more compact and well-separated feature embeddings, leading to improved accuracy. Additionally, techniques such as RetinaFace for detection and embedding optimization strategies further enhance system performance.

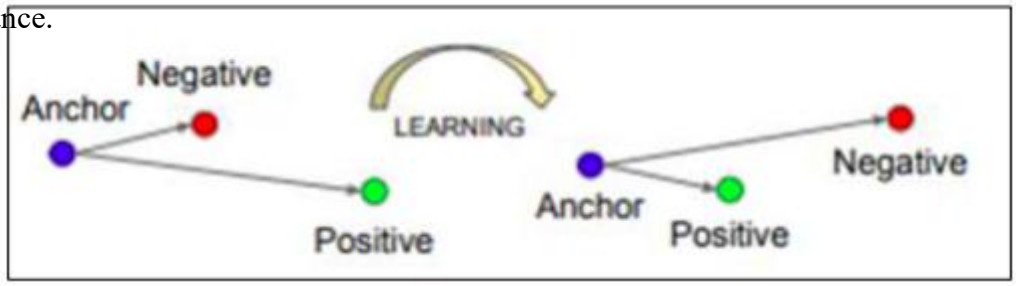

Figure 3 The Triplet Loss

Overall, the literature indicates a clear shift from traditional algorithms to deep learning-based approaches, with models like FaceNet, DeepFace, and ArcFace leading the field. These advancements provide the foundation for developing a robust, accurate, and scalable automated attendance system.

Chapter 3

System Overview

3.1 System Analysis

The proposed system is designed as a *web-based automated attendance platform* that integrates deep learning-based facial recognition with real-time processing. The system replaces traditional attendance mechanisms with an intelligent, contactless, and scalable solution.

It operates through multiple modules, including user registration, facial data processing, real-time recognition, attendance management, and leave handling. The system ensures accurate identity verification while maintaining efficiency and usability.

3.1.1 Functional Requirements

The functional requirements define the core operations that the system must perform. These are essential features required for the system to function effectively.

The system should:

- Allow student registration with name and unique ID
- Capture and store multiple facial images during registration
- Generate and store facial embeddings for each user
- Provide a secure login system for administrators
- Detect and recognize faces in real-time using a camera feed
- Automatically mark attendance upon successful recognition
- Prevent duplicate attendance entries for the same session/day
- Store attendance records with date and timestamp
- Allow retrieval of attendance history and reports
- Implement leave management based on attendance percentage
- Automatically approve or flag leave requests based on predefined criteria

3.1.2 Non-Functional Requirements

Non-functional requirements define the quality and performance aspects of the system.

The system should ensure:

- **Security:** Secure handling of user data and facial embeddings
- **Performance:** Real-time face recognition with minimal delay
- **Accuracy:** High recognition accuracy using deep learning models
- **Scalability:** Ability to handle multiple users simultaneously
- **Usability:** Simple and intuitive user interface
- **Reliability:** Consistent performance under varying conditions
- **Maintainability:** Easy updates and system improvements

3.2 Hardware Requirements

The system requires moderate hardware capabilities to ensure smooth execution of deep learning models and real-time video processing.

Minimum hardware specifications include:

- **Processor:** Intel i5 or higher
- **RAM:** Minimum 16 GB
- **Storage:** 512 GB hard disk or SSD
- **Camera:** Webcam or external camera for face capture

These specifications ensure efficient processing of facial recognition tasks and database operations.

3.4 Tools and Technologies

The development of the system utilizes a combination of computer vision, deep learning, and web technologies.

Key tools and technologies used are:

- **Python (Flask):** Handles backend logic and API development
- **OpenCV:** Used for real-time face detection and image preprocessing
- **DeepFace:** Implements facial recognition using models like ArcFace.
- **PostgreSQL (Neon):** Stores structured attendance and user data
- **Cloudinary:** Manages and stores captured facial images
- **HTML, CSS, JavaScript:** Used for frontend interface design

These technologies collectively enable the system to function efficiently and accurately.

Chapter 4

System Design

4.1 System Architecture

The proposed system follows a *modular web-based architecture* that integrates frontend, backend, and AI-based facial recognition components. The system is designed to ensure scalability, maintainability, and real-time performance.

The frontend interface allows users to interact with the system, while the Flask backend processes requests, performs facial recognition, and communicates with the database.

The architecture is divided into the following components:

- **Frontend Layer:** Built using HTML, CSS, and JavaScript for user interaction
- **Backend Layer:** Flask-based REST API handling logic and processing
- **AI Module:** DeepFace (ArcFace) for facial recognition
- **Database Layer:** PostgreSQL (Neon) for storing user and attendance data
- **Cloud Storage:** Cloudinary for storing facial images

Additionally, the backend is structured using *Flask Blueprints,* which modularize the application into components such as authentication, attendance, and leave management.

4.2 Data Flow Diagram (DFD)

4.2.1 Level-0 DFD

The Level-0 Data Flow Diagram provides a high-level overview of the system.

User → Application → Database

- The user interacts with the application through the web interface
- The application processes input (registration, attendance, leave requests)
- Data is stored and retrieved from the database

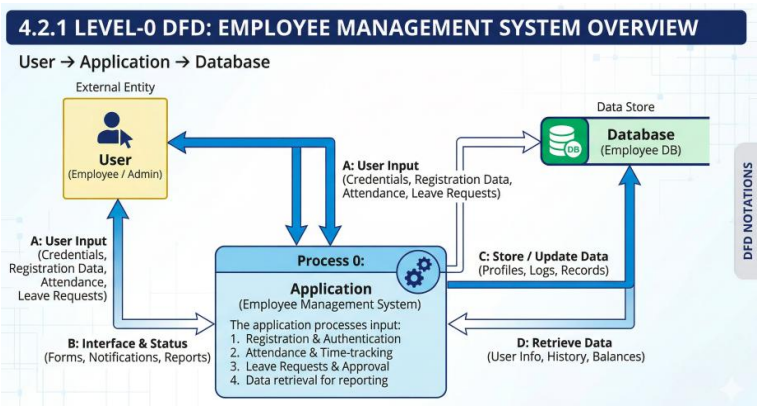

Figure 4 Level- 0 DFD

4.2.2 Level-1 DFD

The Level-1 DFD breaks the system into major modules:

- **User Module:** Handles registration and login
- **Face Processing Module:** Captures images and generates embeddings
- **Attendance Module:** Marks and stores attendance
- **Leave Management Module:** Handles leave requests and approvals
- **Database Module:** Stores user, attendance, and embedding data

Data flows between these modules through API calls and database queries, ensuring smooth system operation.

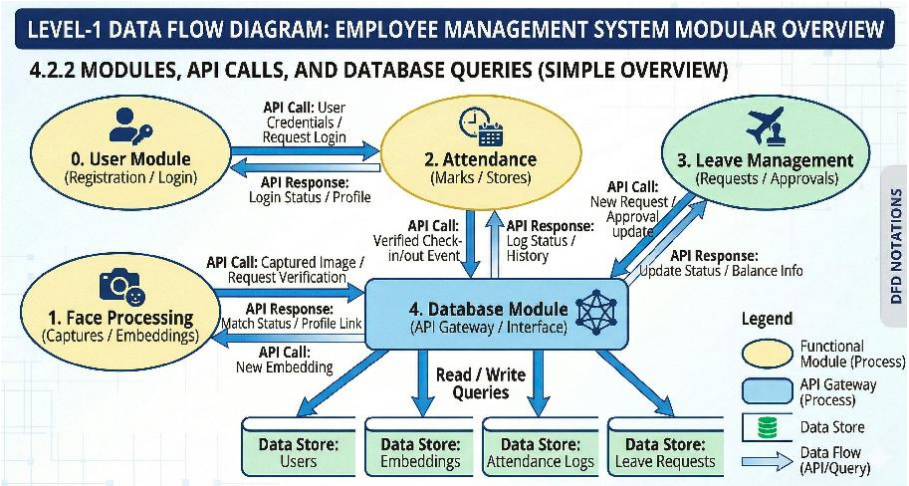

Figure 5 Level-1 DFD

4.3 Use Case Diagram

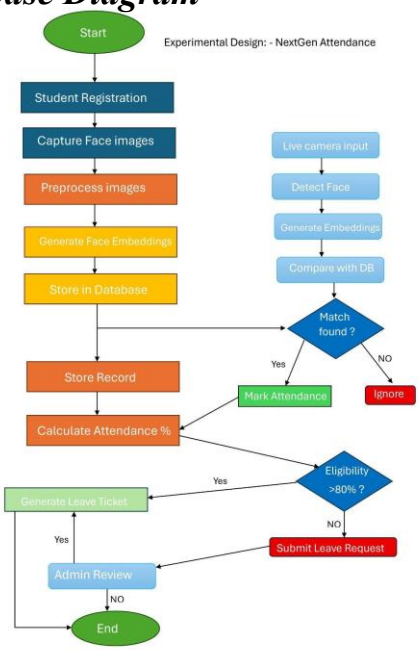

Figure 6 System Architecture flow diagram

The use case diagram illustrates interactions between users and system functionalities, ensuring clarity in system behavior.

4.4 Database Design

The system uses a relational database (PostgreSQL) to store structured data related to users, attendance, and leave management.

Entity-Relationship Overview

The main entities in the system include:

Table Name	Description
Users	Stores student details such as ID, name, and embeddings
Attendance	Records attendance with date and timestamp
Leave	Stores leave requests and approval status

Table 1 Table Information of the real-time database.

4.4.1 Key Relationships

- One **User** can have multiple **Attendance records**
- One **User** can submit multiple **Leave requests**
- Attendance is linked to users via unique ID

4.5 Methodology

The system operates through multiple phases, each responsible for a specific function in the attendance pipeline.

4.5.1 Registration Phase

In this phase, students register by providing their name and unique ID. Multiple facial images are captured using a camera. These images are preprocessed using OpenCV techniques such as face detection and resizing.

Facial embeddings are then generated using DeepFace (ArcFace model), which converts facial features into numerical vectors. These embeddings, along with user details, are stored in the database.

This phase is implemented using:

- *capture_face()* → Captures dataset
- *generate_embeddings()* → Generates embeddings

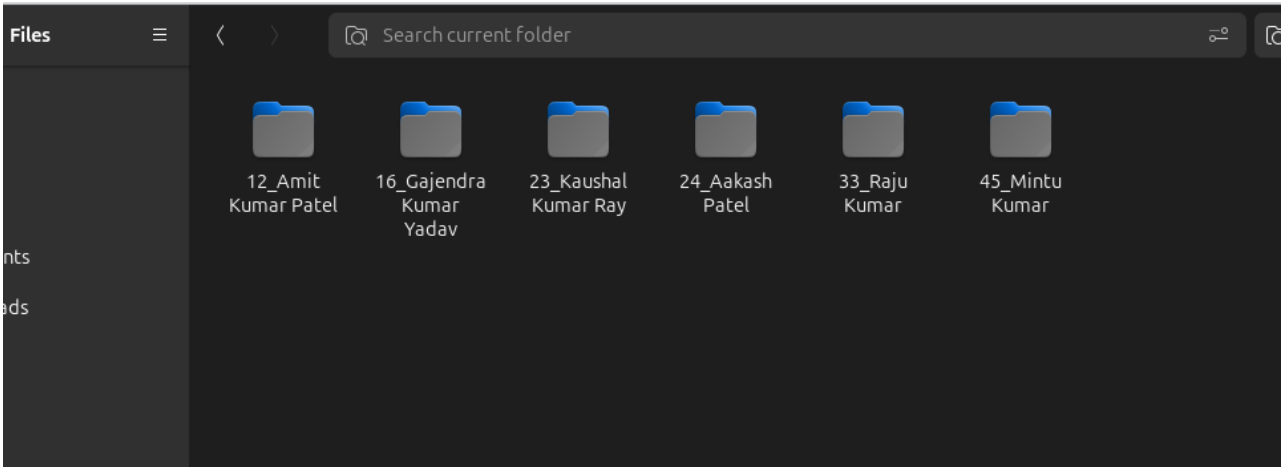

Figure 7 A Snapshot of face database

Figure 8 Detected and cropped faces

4.5.2 Training Phase

In the training phase, collected images are processed to generate embeddings for all registered users. These embeddings are stored in a serialized file (*embeddings.pkl*) for efficient access during recognition.

This step prepares the system for real-time face matching.

Implemented using: *train_model()*

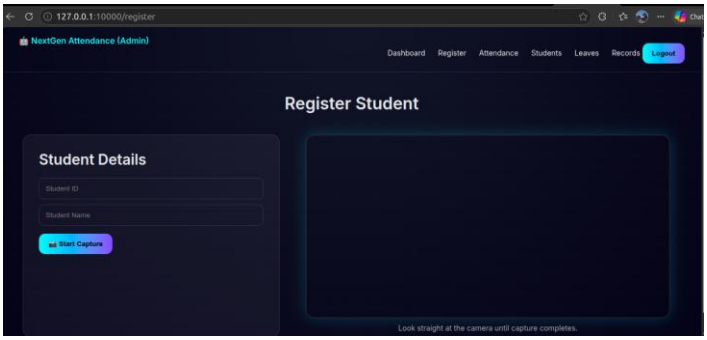

Figure 9 Student Registration and Training model

4.5.3 Attendance Phase

During this phase, the system captures live video input through a camera. Faces are detected using OpenCV, and embeddings are generated in real-time.

These embeddings are compared with stored embeddings using similarity metrics such as cosine distance. If a match is found, the system identifies the user and marks attendance automatically.

Implemented using: *process_attendance()*

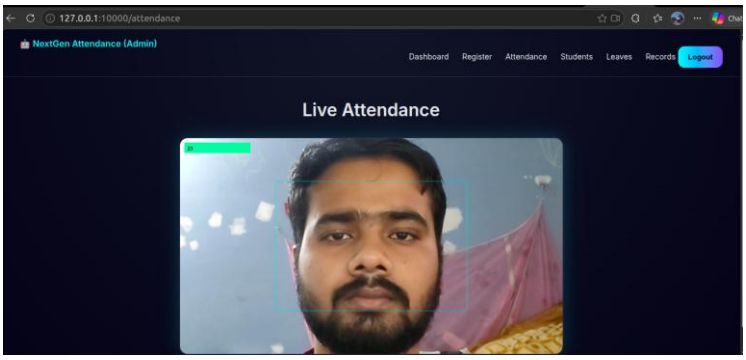

Figure 10 Marking live attendance

4.5.4 Attendance Logic

The system ensures accurate attendance management through the following logic:

- Attendance is stored with date and timestamp
- Duplicate entries are prevented for the same day
- Attendance history can be retrieved
- Attendance percentage is calculated dynamically

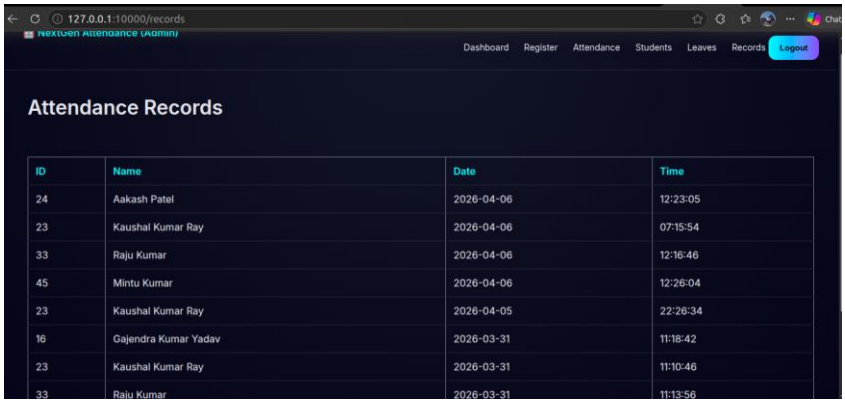

Figure 11 Retrieval of the stored attendance from the database

4.5.5 Leave Management System

The leave management module is integrated with attendance data. Leave approval is based on attendance percentage:

- `If attendance ≥ 80% → Automatically approved`
- If attendance < 80% → Marked as pending

Implemented in: *leave_routes.py*

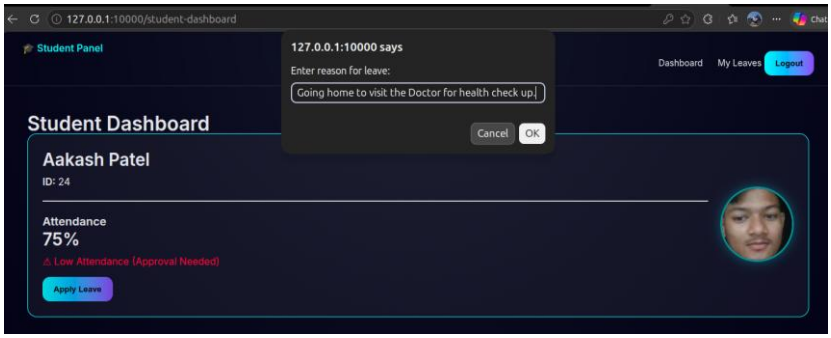

Figure 12 Retrieval of the stored attendance from the database.

Chapter 5

System Implementation

5.1 Frontend Implementation

The frontend of the system is developed using HTML5, CSS, and JavaScript, providing a simple and user-friendly interface for interaction. The design focuses on usability, clarity, and real-time feedback.

The frontend is responsible for handling user interactions such as registration, attendance marking, and viewing attendance records. It communicates with the backend through HTTP requests (REST APIs).

Key features of the frontend include:

- User Interface Design: Clean layout for registration, attendance, and dashboard pages
- Form Handling: Input fields for student registration (name, ID)
- Camera Integration: Interface for capturing facial images using a webcam
- Real-Time Feedback: Display messages for successful attendance marking or errors
- Responsive Design: Ensures usability across different devices

CSS is used to enhance visual appearance, improve readability, and ensure proper alignment of elements. JavaScript handles dynamic behavior such as capturing images, sending requests, and updating UI elements.

5.2 Backend Implementation

The backend is implemented using Python with the Flask framework, which manages application logic, API routing, and integration with AI models and the database.

Flask is structured using Blueprints, allowing modular separation of components such as authentication, attendance, and leave management.

The backend performs the following operations:

- Handles API requests from the frontend
- Processes facial recognition using DeepFace

- Manages attendance logic and validation
- Interacts with PostgreSQL database
- Integrates with Cloudinary for image storage

Key Backend Components

- Flask Application: Core server handling requests and responses
- Blueprints: Modular structure for routes (auth, attendance, leave)
- REST APIs: Enable communication between frontend and backend
- DeepFace Integration: Generates facial embeddings and performs matching
- OpenCV: Handles real-time face detection and preprocessing

Example functional modules:

- *capture_face()* → Captures user facial images
- *generate_embeddings()* → Converts images into embeddings
- *process_attendance()* → Matches faces and marks attendance
- *leave_routes.py* → Handles leave logic and approval system

5.3 Database Implementation

The system uses *PostgreSQL (Neon cloud database)* to store structured data. The database ensures efficient storage, retrieval, and management of user and attendance records

Field Name	Data Type	Description
Id	VARCHAR	Unique student ID (Primary Key)
name	VARCHAR	Name of the Student
Created_at	TIMESTAMP	Timestamp when the record was created (default: current time)

.**Table 2** Students Records Storage

Field Name	Data Type	Description
Id	SERIAL	Unique attendance record ID (Primary Key)
Student_id	VARCHAR	References student id (foreign Key)
date	DATE	Date of attendance
time	TIME	Time of attendance marking

Table 3 Architecture of the attendance marking section

Field Name	Data Type	Description
Id	SERIAL	Unique leave request ID (Primary Key)
Student_id	VARCHAR	References student ID
Reason	TEXT	Reason for leave
Status	VARCHAR	Leave status (default: 'Pending')
Created_at	TIMESTAMP	Timestamp of leave request

Table 4 Leave request management

5.3.1 Database Operations

The backend performs various database operations such as:

- **Insertion:** Adding new users and attendance records
- **Retrieval:** Fetching attendance history and user details
- **Update:** Modifying leave status (approved/pending)
- **Validation:** Checking duplicate attendance entries

Example logic:

- Attendance is marked only if no entry exists for the current date
- Attendance percentage is calculated dynamically using stored records

5.4 Key Features Implementation

The system includes several important features implemented using a combination of deep learning and web technologies.

5.4.1 Face Recognition-Based Attendance

The core feature of the system is automated attendance marking using facial recognition.

- Faces are detected using OpenCV
- DeepFace generates embeddings using ArcFace/FaceNet models
- Embeddings are compared with stored data using similarity metrics
- Attendance is marked automatically upon successful match

```
127.0.0.1 - - [06/Apr/2026 18:13:21] "POST /process_attendance HTTP/1.1" 200 -
Distance: 2.6926543110909673
detected: 23
127.0.0.1 - - [06/Apr/2026 18:13:25] "GET /students HTTP/1.1" 200 -
127.0.0.1 - - [06/Apr/2026 18:13:26] "GET /static/css/style.css HTTP/1.1" 304 -
[DB SUCCESS] 23
```

Figure 13 Attendance marking and updating in database (Neon).

5.4.2 Automated Attendance Management

The system ensures accurate attendance tracking through:

- Timestamp-based attendance recording
- Prevention of duplicate entries
- Retrieval of attendance history
- Automatic calculation of attendance percentage

5.4.3 Leave Management System

A rule-based leave management system is implemented:

- Attendance percentage is calculated
- `If ≥ 80% → Leave is auto-approved`
- If < 80% → Leave remains pending

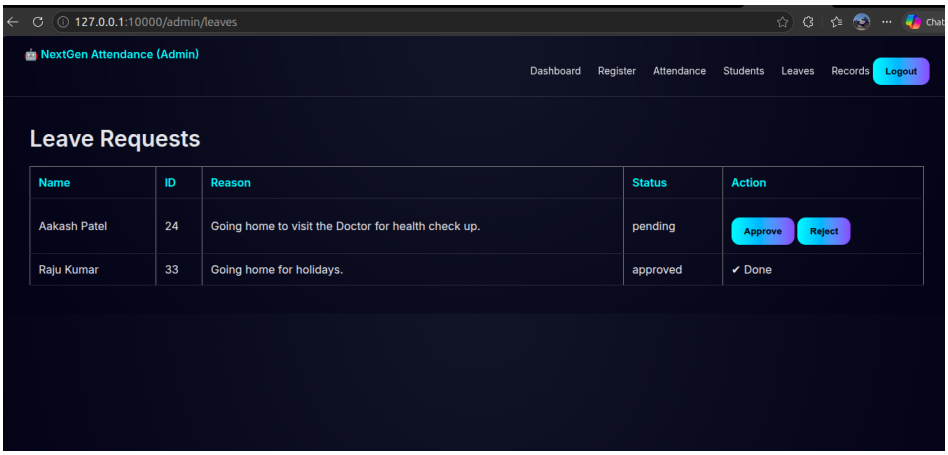

Figure 14 Decision making dashboard leave requests.

This reduces administrative workload and ensures fairness. Also, this architecture enables sustainable implementation of the modern systems with less overheads.

5.4.4 Cloud-Based Image Storage

The system uses *Cloudinary* to store captured facial images:

- Images are uploaded securely to the cloud
- Reduces local storage dependency
- Enables scalability and easy access

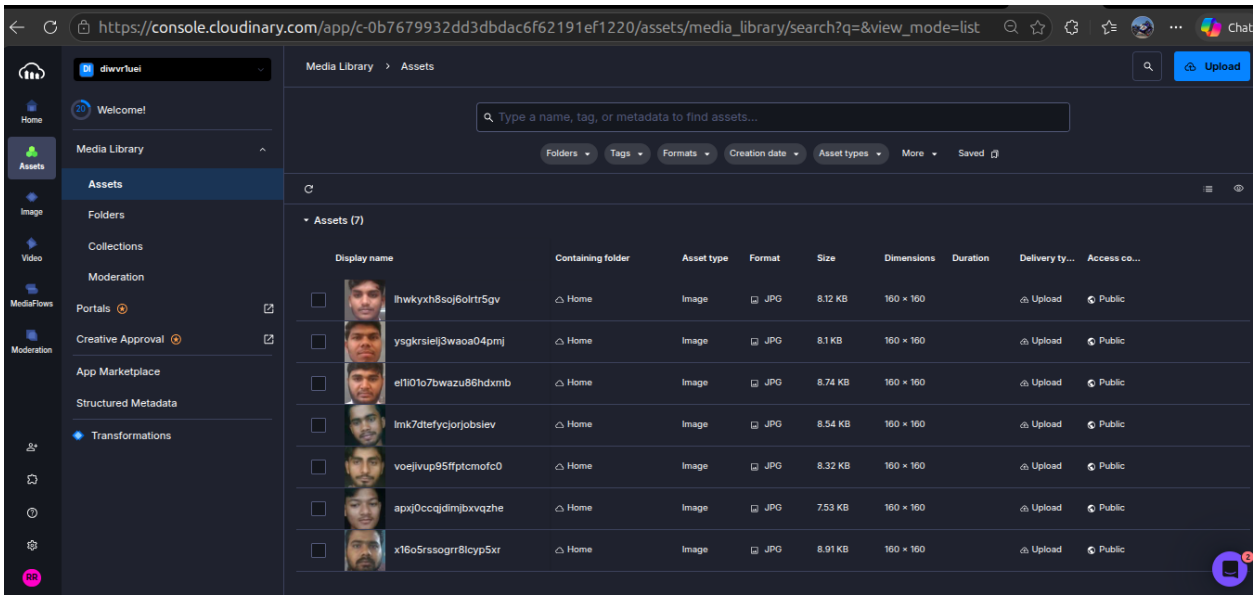

Figure 15 Cloudinary assets dashboard where the facial data are stored.

5.4.5 Real-Time Processing

The system supports real-time face detection and recognition:

- Live camera feed processing
- Instant embedding generation
- Fast comparison with stored data
- Immediate attendance marking

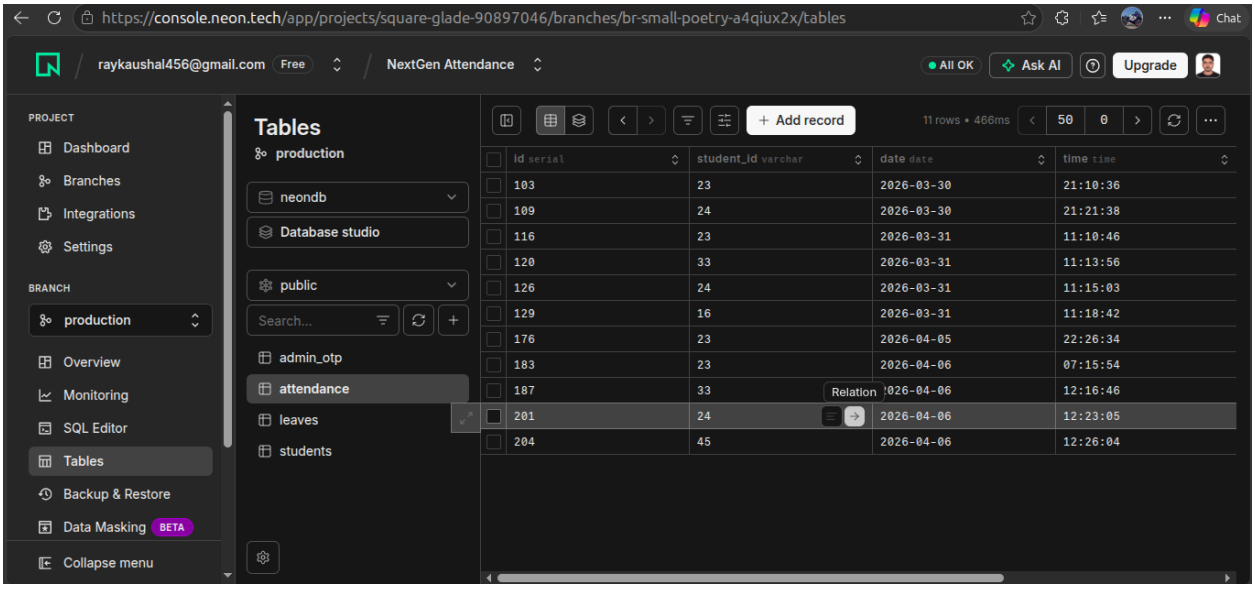

Figure 16 Neon Database for real-time manipulation of the data.

5.4.6 Modular Backend Design

The use of Flask Blueprints ensures:

- Clean code organization
- Easy debugging and maintenance
- Scalability for adding new features

Chapter 6
System Testing

6.1 Introduction to Testing

System testing is a critical phase in software development that ensures the application functions correctly, efficiently, and reliably under different conditions. The proposed attendance system was tested to verify its functionality, accuracy, performance, and robustness.

Testing was performed on different modules including user registration, face recognition, attendance marking, and leave management. Both functional and performance testing techniques were applied to ensure that the system meets the required specifications.

6.2 Test Cases

Test cases were designed to validate the correctness of individual functionalities. Each test case includes input conditions and expected outputs.

Test Case	Input	Expected Output
User Registration	Valid name and ID	User successfully registered
Face Capture	Camera enabled	Facial images captured and stored
Embedding Generation	Valid images	Embeddings generated successfully
User Login (Admin)	Correct credentials	Login successful
User Login (Admin)	Incorrect credentials	Error message displayed
Face Recognition	Registered face	User identified correctly
Face Recognition	Unregistered face	Access denied / no match
Attendance Marking	Recognized face	Attendance recorded with timestamp

Duplicate Attendance Check	Same user same day	Duplicate entry prevented
Leave Request	`Attendance ≥ 80%`	Leave auto-approved
Leave Request	Attendance < 80%	Leave marked as pending
Database Storage	Valid data	Data stored correctly in PostgreSQL

Table 5 Functional Test Cases

6.3 Performance Testing

Performance testing was conducted to evaluate the system's efficiency, responsiveness, and scalability under different conditions.

The following parameters were tested:

- **System Response Time:**
  The time taken for face detection, recognition, and attendance marking was measured. The system demonstrated near real-time performance with minimal delay.
- **Load Handling:**
  The system was tested with multiple users and repeated requests to evaluate stability. The backend handled concurrent requests efficiently without significant performance degradation.
- **Processing Speed:**
  The time required to generate embeddings and compare them with stored data was optimized using precomputed embeddings (*embeddings.pkl*).
- **Database Performance:**
  PostgreSQL queries were tested for speed and reliability in storing and retrieving attendance records.

6.4 Testing Outcomes

The testing phase confirmed that:

- The system correctly identifies registered users with high accuracy
- Attendance is recorded accurately with proper timestamp validation
- Duplicate entries are effectively prevented
- Leave management logic works as expected

- The system performs efficiently under normal and moderate load conditions

Minor issues related to performance and accuracy were identified and resolved during development, leading to improved system reliability.

Chapter 7
Results and Discussion

7.1 Overview of Results

The developed system was successfully implemented and tested for real-time attendance management using facial recognition. The results demonstrate that the system is capable of accurately detecting and recognizing faces, marking attendance automatically, and managing attendance records efficiently.

The integration of deep learning models significantly improved recognition accuracy compared to traditional methods. The system performs reliably under normal conditions and provides real-time responses with minimal delay.

7.2 System Outputs

The system generates multiple outputs during operation, reflecting different functionalities of the application.

Key Outputs Include:

- User Registration Output:
  Displays confirmation after successful registration and embedding generation
- Face Recognition Output:
  Identifies the user in real-time and displays the recognized name/ID
- Attendance Output:
  Automatically records attendance with date and timestamp
- Leave Management Output:
  Displays leave approval or pending status based on attendance percentage
- Dashboard Output:
  Shows attendance history and summary for users/admin

7.3 Screenshots and System Interface

The system interface was designed to be simple and intuitive, ensuring ease of use for both students and administrators.

Major Screens in the System:

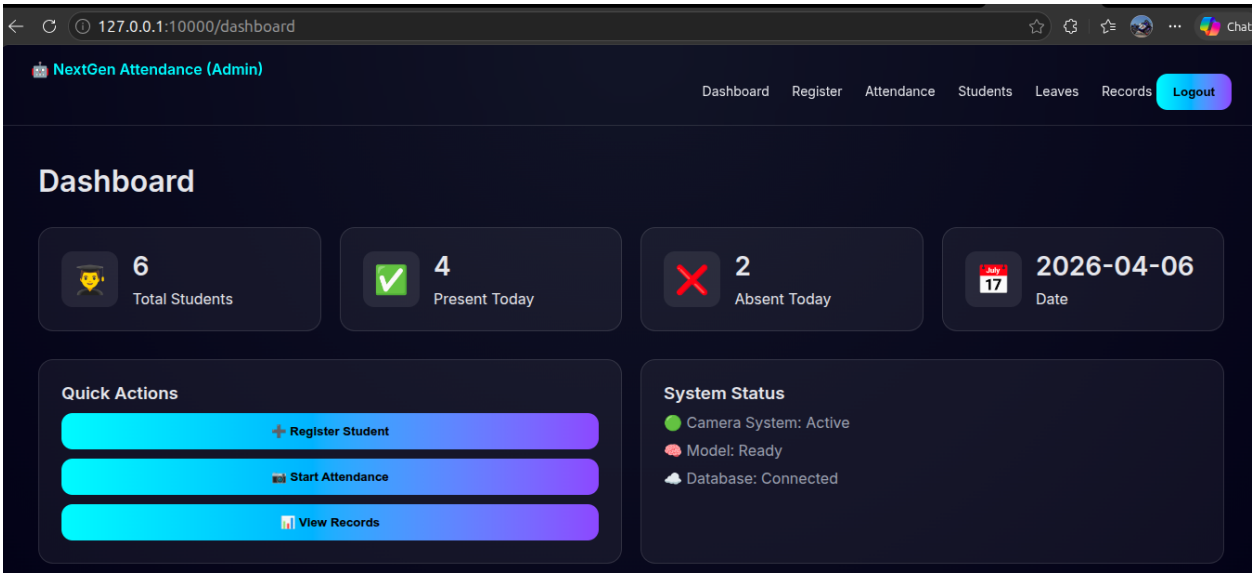

Figure 17 Homepage Dashboard.

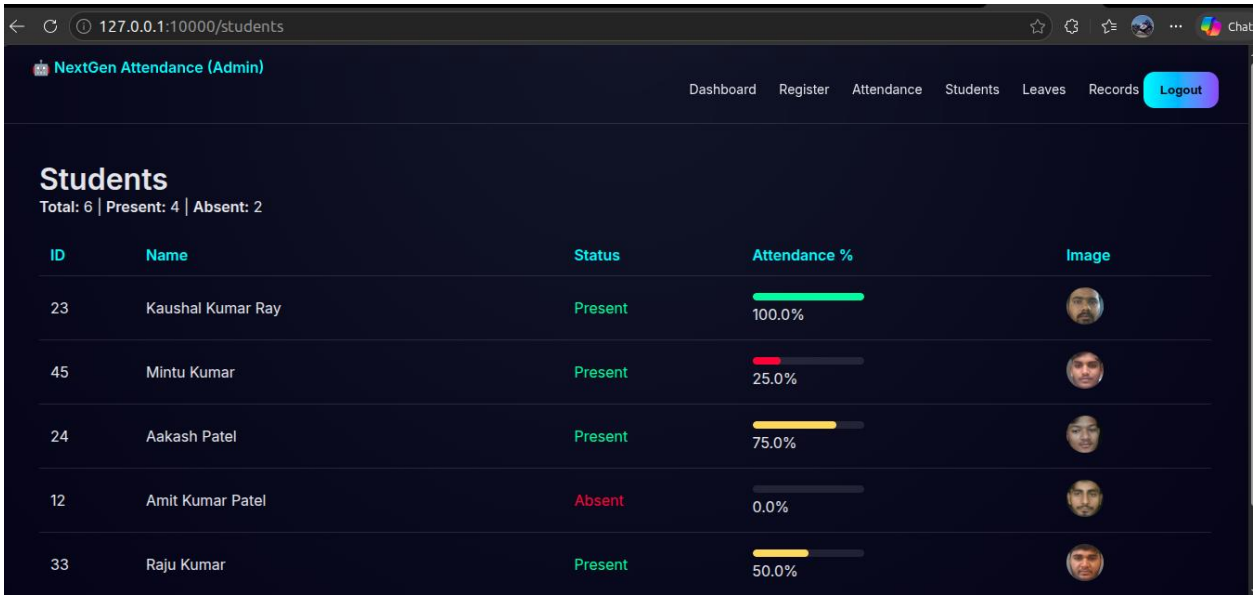

Figure 18 Attendance dashboard displaying records.

7.4 Performance Analysis

The system was evaluated based on accuracy, speed, and reliability.

- **Recognition Accuracy:**
  The use of DeepFace with ArcFace significantly improved recognition accuracy. Capturing multiple images per user further enhanced performance.

- **Processing Time:**
  The system performs face recognition and attendance marking in near real-time, ensuring minimal delay during operation.
- **System Reliability:**
  The system consistently identifies users under standard conditions and prevents duplicate attendance entries.
- **Scalability:**
  The system can handle multiple users efficiently, making it suitable for institutional deployment.

7.5 Discussion

The results indicate that the proposed system effectively addresses the limitations of traditional attendance systems. By leveraging deep learning-based facial recognition, the system eliminates manual intervention and reduces the possibility of proxy attendance.

The use of embedding-based models ensures robust performance even with variations in facial appearance. However, certain challenges such as lighting conditions and occlusions can still impact accuracy, which can be further improved with enhanced datasets and model tuning.

Overall, the system demonstrates strong potential for real-world deployment in educational institutions and organizations.

7.6 Summary

The system successfully achieves its objectives by:

- Automating attendance using facial recognition
- Improving accuracy and efficiency
- Reducing manual effort and proxy attendance
- Providing a scalable and user-friendly solution

Chapter 8

Conclusion and Future Work

8.1 Conclusion

The proposed NextGen Attendance System provides an efficient, intelligent, and automated solution for attendance management using deep learning-based facial recognition. The system successfully eliminates the need for manual attendance marking and significantly reduces the possibility of proxy attendance.

The implementation of facial recognition using advanced models such as ArcFace and FaceNet enables accurate identification of individuals in real-time. The system effectively captures facial data, generates embeddings, and compares them with stored records to mark attendance automatically.

During development, several challenges were encountered, including incorrect attendance logic, database connectivity issues, low recognition accuracy, UI limitations, and performance bottlenecks. These challenges were systematically addressed through improvements such as date-based attendance validation, optimized backend processing, enhanced UI design, and the use of more robust deep learning models.

The final system demonstrates:

- Accurate real-time face detection and recognition
- Automated attendance marking with timestamp validation
- Improved accuracy using deep learning and data augmentation
- Efficient database management and record handling
- A user-friendly web interface

Overall, the system proves to be scalable, reliable, and suitable for deployment in educational institutions and organizations, offering a modern alternative to traditional attendance systems.

8.2 Future Work

Although the system performs effectively, several enhancements can be implemented to further improve its capabilities and usability.

Future scope includes:

- **Mobile Application Integration:**
  Developing a mobile-based version of the system for increased accessibility
- **Advanced Security Features:**
  Implementing liveness detection to prevent spoofing using photos or videos
- **Improved Accuracy:**
  Training models on larger and more diverse datasets to handle extreme conditions
- **Cloud Deployment:**
  Deploying the system on cloud infrastructure for better scalability and remote access
- **Real-Time Analytics Dashboard:**
  Adding graphical insights for attendance trends and performance tracking
- **Multi-Camera Support:**
  Enabling simultaneous attendance tracking across multiple classrooms
- **Integration with Institutional Systems:**
  Connecting with ERP or academic systems for centralized data management

References

- Deng et al., *ArcFace: Additive Angular Margin Loss for Deep Face Recognition*
- Essien et al., *Deep Learning-Based Face Recognition Attendance System* (GJETA)
- Schroff et al., *FaceNet: A Unified Embedding for Face Recognition*

Appendix

(Project link : https://github.com/Kaushal-Kumar-Ray/NextGen_Attendance)